

Expanding Chemical Space in the Synthesis of Gold Bipyramids

Ana Sánchez-Iglesias and Marek Grzelczak*

Gold bipyramids (AuBPs), despite having superior properties compared to their spectroscopically similar counterparts, gold nanorods, have found comparatively limited applications. This discrepancy is primarily due to the lack of protocols to tailor their dimensions. Typically, the concentration of Au seeds is virtually the sole factor that determines the aspect ratio and thus, the optical properties of AuBPs. As a result, varying the volumes of AuBPs while incurring minimal changes to their optical spectra remains a synthetically non-trivial task. Here, the chemical space in the seeded growth of AuBPs, is expanded by exploiting the interplay between bromide, silver ions, and seed concentration for tuning the final dimensions and optical properties of AuBPs. Specifically, a 6 fold change in volumes of AuBPs is achieved while maintaining the fixed plasmon band position. Further overgrowth of as-prepared bipyramids broadens the realizable dimensions without compromising quality and initial morphology. Overall, the results expand the chemical toolbox in the wet-chemistry synthesis of anisotropic gold nanoparticles, which is relevant for health, colorimetric sensors, and energy applications.

thermal stability of AuBPs, a primary requisite for thermoplasmonic transducers,^[18–20] photothermal therapy,^[21–23] and photocatalysis.^[24–26] Also, AuBPs are attractive building blocks in directed self-assembly, featuring unique optical properties.^[27,28] The application rate of AuBPs in various fields goes hand in hand with the progress in synthetic methods for their preparation. Over the last two decades, we witnessed major breakthroughs in the synthesis of gold bipyramids involving i) the need for pentatwinned seeds to render bipyramid morphology,^[29] ii) reversible silver overgrowth to increase nanocrystals yield through post-synthetic purification,^[7,30] iii) the use of hydroxyquinoline as an alternative reducing agent,^[31] iv) thermal treatment of seeds^[31,32] to increase the yield (85%) without the need for post-synthetic purification.^[32] These protocols remain highly constrained, that is, controlling the volume while keeping a constant aspect ratio is synthetically

1. Introduction

Gold bipyramids (AuBPs) are gaining significant popularity as a class of anisotropic nanocrystals due to their superior plasmonic performance compared to gold nanorods.^[1,2] Due to low radiative damping, the AuBPs feature exceptional narrow plasmon linewidth,^[3,4] large local electric field enhancements at the sharp tips,^[5,6] rendering them an appealing platform for sensors,^[7–12] imaging devices,^[13,14] coupled^[15,16] and nonlinear^[17] hybrid systems. The penta-twinned crystalline structure improves the

challenging, a feature that is central for modulating scattering-to-extinction ratio,^[33] which, on the other hand, determines the targeted applications.^[34] Thus, there is an urgent need for expanding the chemical space in the synthesis of gold bipyramids to unlock their dimensional and optical diversity, as has been also indicated by data-driven analysis of protocols available in the literature.^[35,36]

The chemical space of gold bipyramids encompasses six key parameters: surfactant, gold precursor, silver ions, reducing agent, pH, and seeds, with the seed concentration playing a pivotal role in determining the aspect ratio. Nevertheless, adjusting the seed concentration simultaneously impacts the final nanocrystal volume, complicating systematic investigations into volume-dependent plasmonic properties, such as photo-thermal conversion and^[37,38] plasmon-exciton coupling.^[15,16] Parola and co-workers^[31] have attempted to address this challenge by examining the influence of surfactant counterions on the aspect ratio of bipyramids. Their findings revealed that bromide (Br^-) in cetyltrimethylammonium (CTA^+) leads to nanocrystals with low aspect ratios, whereas a mixture of bromide and chloride (Br^-/Cl^-) in CTA^+ promotes the formation of elongated bipyramids, termed nanojavelins. Furthermore, the same group demonstrated that varying CTAB concentrations in a single-surfactant system can suppress the production of isotropic nanoparticles.^[39] These results suggest that the role of

A. Sánchez-Iglesias, M. Grzelczak
Centro de Física de Materiales (CSIC-UPV/EHU)
Paseo Manuel de Lardizabal 5, Donostia-San Sebastián 20018, Spain
E-mail: marek.g@csic.es
M. Grzelczak
Donostia International Physics Center (DIPC)
Paseo Manuel de Lardizabal 4, Donostia-Sebastián 20018, Spain

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202407735>

© 2024 The Author(s). Small published by Wiley-VCH GmbH. This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](#) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

DOI: 10.1002/smll.202407735

counterion interactions within surfactant systems could serve as an additional parameter in controlling the dimensions of gold bipyramids. However, the practical implementation of this approach remains a challenge.

This work deals with expanding the structural parameters of AuBPs through fine-tuning of synthetic parameters. The main finding is that CTAB as an additive (being CTAC the primary surfactant), can tailor the aspect ratio of AuBPs at fixed volume. This is in stark contrast to the state-of-the-art, where similar modulations of localized surface plasmon resonance (LSPR) maxima required varying the amount of Au seeds, and ultimately changing the volumes of final AuBPs.^[32] Kinetic studies showed that the bipyramidal morphology is favored under a fast reduction rate (low CTAB concentration), indicating that the growth pathway of AuBPs is a subtle interplay between kinetic control and surface passivation. Accordingly, we showed that silver nitrate, although critical for symmetry breaking, has a minor effect on tailoring the aspect ratio of final bipyramids. Furthermore, we found that by fine-tuning seeds and CTAB concentrations, one can obtain AuBPs of similar plasmon band position (e.g., 750 nm) and a 6 fold change in volume. The overgrowth of the as-prepared bipyramids allows for expanding available dimensions while conserving the original shape.

2. Results and Discussion

This work builds on the previous findings in which Bromide (Br^-) ions induce symmetry breaking to render anisotropic nanoparticles (nanorods) in the presence of hexadecyltrimethylammonium chloride (CTAC) as the primary surfactant,^[40,41] where Br^- ions are responsible for preferential adsorption of CTA^+ molecules on high-energy gold facets^[42–45] and alter the reduction potential of gold ions,^[46] favoring thus anisotropic growth. We hypothesized that by modulating CTAB while keeping the concentration of CTAC as the primary surfactant constant, one could modulate the reduction rate of gold precursor and, thus, the aspect ratio of AuBPs. To evaluate the hypothesis, we implemented a standard seeded-growth model that involves the use of CTAC-stabilized and thermally-aged seed nanocrystals of 6 nm (Videos S1 and S2, Supporting Information)^[32] that are added to the growth solution comprising the cationic surfactants (CTAC and CTAB), silver nitrate, and 8-hydroxyquinoline (8-HQL) as mild reducing agent at 35 °C. Note that the use of 8-HQL, instead of ascorbic acid, does not require an extra HCl in the growth solution to minimize the formation of isotropic byproduct.^[31] We found that the synthesis in pure CTAB (100 mM) renders rice-like bipyramids, while in pure CTAC (100 mM), bipyramids and isotropic byproducts (Figure 1b; Figure S1, Supporting Information). To evaluate the effect of Br^- concentration on AuBP aspect ratio, we synthesized 13 samples by keeping constant CTAC concentration (100 mM) while modulating CTAB concentration from 1 to 50 mM, keeping all other experimental parameters constant (Table S1 and Videos S3 and S4, Supporting Information). We observed that with increasing CTAB concentration, the plasmon band position blueshifts from 1070 nm (1 mM CTAB) to 584 nm (50 mM CTAB) (Figure 1c,i). Accordingly, the structural analysis by transmission electron microscopy (TEM) confirmed a decrease in length from 100 ± 3 nm to 49 ± 4 nm, an increase in width from 17 ± 1 to 29

± 1 nm, and thus decreasing the aspect ratio from 5.9 ± 0.4 to 1.7 ± 0.2 with the increase of CTAB concentration from 1 to 50 mM (Figure 1a,f,h; Figures S2 and S3 and Table S1, Supporting Information). Importantly, keeping constant the seeds number, we could obtain AuBPs of the volume falling in a narrow range of $\approx 11\,000\text{ nm}^3$ (Figure 1g). The critical role of bromide ions in the modulation of aspect ratio was confirmed by using pure KBr instead of CTAB (Figure S4 and Table S2, Supporting Information), where LSPR blueshifts with increasing bromide concentration.

It is commonly agreed that the slower the reduction kinetics, the larger the shape anisotropy, especially for gold nanorods.^[40,47–49] Intuitively, one expects to obtain AuBPs of a higher aspect ratio by slowing the reduction kinetics. We performed kinetic studies for all 13 samples in Figure 1 ([CTAB] = 1 – 50 mM) in the presence of CTAC (100 mM), and $\text{Au}^{3+}:\text{Au}^0 = 67$ molar ratio (Figure 1d,e; Figure S5, Supporting Information). At the lowest CTAB concentration (1mM), the gold salt reduction was completed in ≈ 30 min, whereas at 50 mM CTAB, it required 350 min (Figure 1d,e). With increasing Br^- ions (slowing kinetics), the maximum of the LSPR plasmon band decreased from 1030 to 584 nm (Figure 1e): the faster the reduction, the larger the aspect ratio, indicating that the growth mechanism of pentatwinned bipyramids is inherently different from that of single crystal gold nanorods^[50] and involves an interplay of both kinetic and surface passivation effects.^[51] Recent studies show that the growth of bipyramids commences from pentatwinned seeds to nanograins to receive a bipyramidal shape.^[29,39,52] The binding affinity of CTAB to the gold surface is greater than that of CTAC in the presence of silver ions through the formation of $\text{CTA}^+/\text{[AgBr}_2\text{]}^-$ that exhibit stronger binding affinity to high index facets as compared to chloride-bearing $\text{CTA}^+/\text{[AgCl}_2\text{]}^-$.^[52] At the early stage of the growth, the formed nanograins feature {112} facets^[39] that are stabilized preferentially by $\text{CTA}^+/\text{[AgBr}_2\text{]}^-$. Thus, at high CTAB concentration, the strong binding of $\text{CTA}^+/\text{[AgBr}_2\text{]}^-$ slows the rate of monomer feeding (slow kinetics) along the twin axis, rendering low aspect ratio bipyramids with larger tip curvature (Figure S3, Supporting Information). At low CTAB concentration, the rate of monomer feeding is high (fast kinetics) since fewer $\text{CTA}^+/\text{[AgBr}_2\text{]}^-$ are available to passivate nanograins {112} facets, resulting in a fast reduction rate along twin axis to render higher aspect ratio bipyramids with sharp tips (Figure S3, Supporting Information).^[53] Additionally, independent of surface events, the variable concentration of Br^- can alter the ability of gold ions to be reduced by 8-HQL because of the difference in reduction potential of AuCl_2 (1.154 V) and AuBr_2 (0.960 V) (Figure S6, Supporting Information).^[51] Therefore, the lower the Br^- concentration (lower reduction potential), the faster the kinetics, leading to preferential growth along the long axis. Overall, we postulate that the growth kinetics of AuBPs resemble the growth mechanism of gold nanostars, featuring a fast metal deposition to render spikes, a characteristic property of pentatwinned nanocrystals.^[54–56]

The concentration of Ag^+ is a primary parameter in controlling aspect ratio in single-crystalline gold nanorods.^[49,57] We observed that such a principle does not hold for AuBPs. We performed experiments increasing silver concentration from 0 to 100 μM while keeping other parameters constant

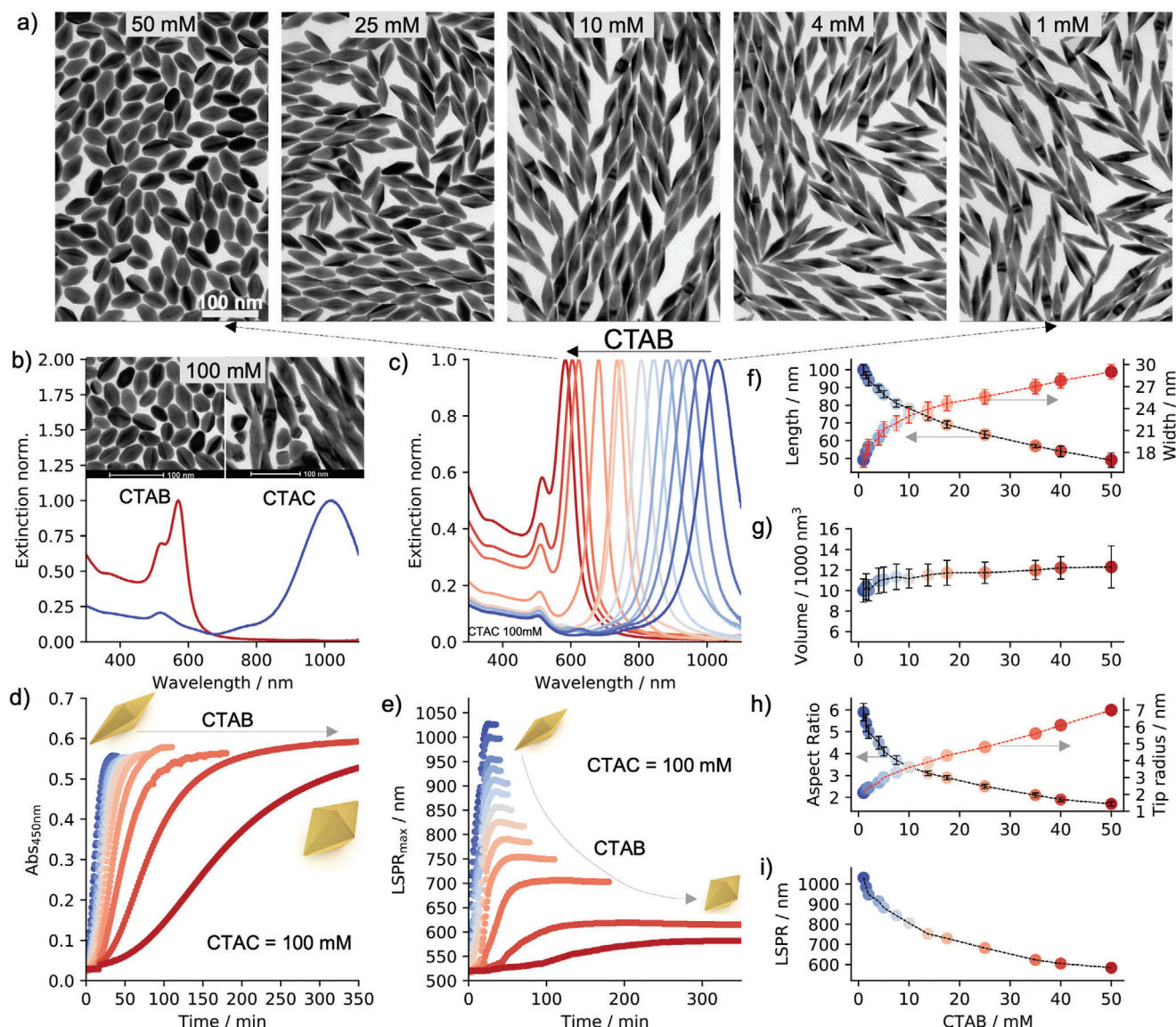


Figure 1. Controlling dimensions of AuBPs through CTAB concentration. a) TEM images of AuBPs obtained with increasing CTAB concentration from 1 to 50 mM at CTAC = 100 mM b) UV-vis-NIR spectra of AuBPs obtained in pure CTAB and CTAC (100 mM), showing different optical and morphological (inset) properties. c) UV-vis-NIR extinction spectra of AuBPs obtained with increasing CTAB concentration from 1 to 50 mM as indicated by the arrow. d, e) Time-dependent evolution of absorbance (450 nm) and LSPR position during the AuBPs growth with increasing CTAB concentration. f–i) Effect of CTAB concentration on length and width (f), volume (g), aspect ratio and tip radius (h), and the position of LSPR (i), showing that volume of AuBPs is the sole invariant structural parameter.

([Gold seeds] = 12.5 μ M, [CTAC] = 100 mM, and [CTAB] = 5 mM) (Table S3, Supporting Information). In the absence of silver ions, the isotropic nanoparticles were the only product, featuring a monomodal plasmon band at 527 nm (Figure 2a,b,d; Figure S7, Supporting Information). The increase of Ag^+ to 23 μ M caused the emergence of a longitudinal plasmon band that progressively redshifted to 730 nm, accompanied by the drop of the population of isotropic byproduct (Figure 2a,b,d,e), corroborating the findings by Xu et al.^[21] A silver concentration of 30 μ M was set as a threshold to ensure high-quality product (98% of AuBPs) (Figure 2e). Interestingly, the silver concentration has a minor impact on the LSPR position and the dimensions at the prod-

uct purity above 95% (Figure 2b,c; Table S3 and Figure S7, Supporting Information). For Ag^+ concentration from 30 to 100 μ M, the LSPR redshifted only from 737 to 753 nm while the length and width values remained within the range of statistical error (Figure 2c). The silver concentration threshold of 30 μ M is related to the total surface final nanoparticles. It has been shown that the gold {100} facet of 100 nm² can accommodate $\approx$ 1000 silver atoms.^[51] The dimensions of AuBPs obtained at Ag^+ = 30 μ M are 48 by 16 nm, giving a total surface of 2.8×10^{16} nm². The supply of 30 μ M of Ag^+ during the growth assures 1290 Ag atoms per 100 nm² of Au. Therefore, as long as sufficient Ag^+ is provided to form a monolayer on growing AuBPs, the symmetry breaking

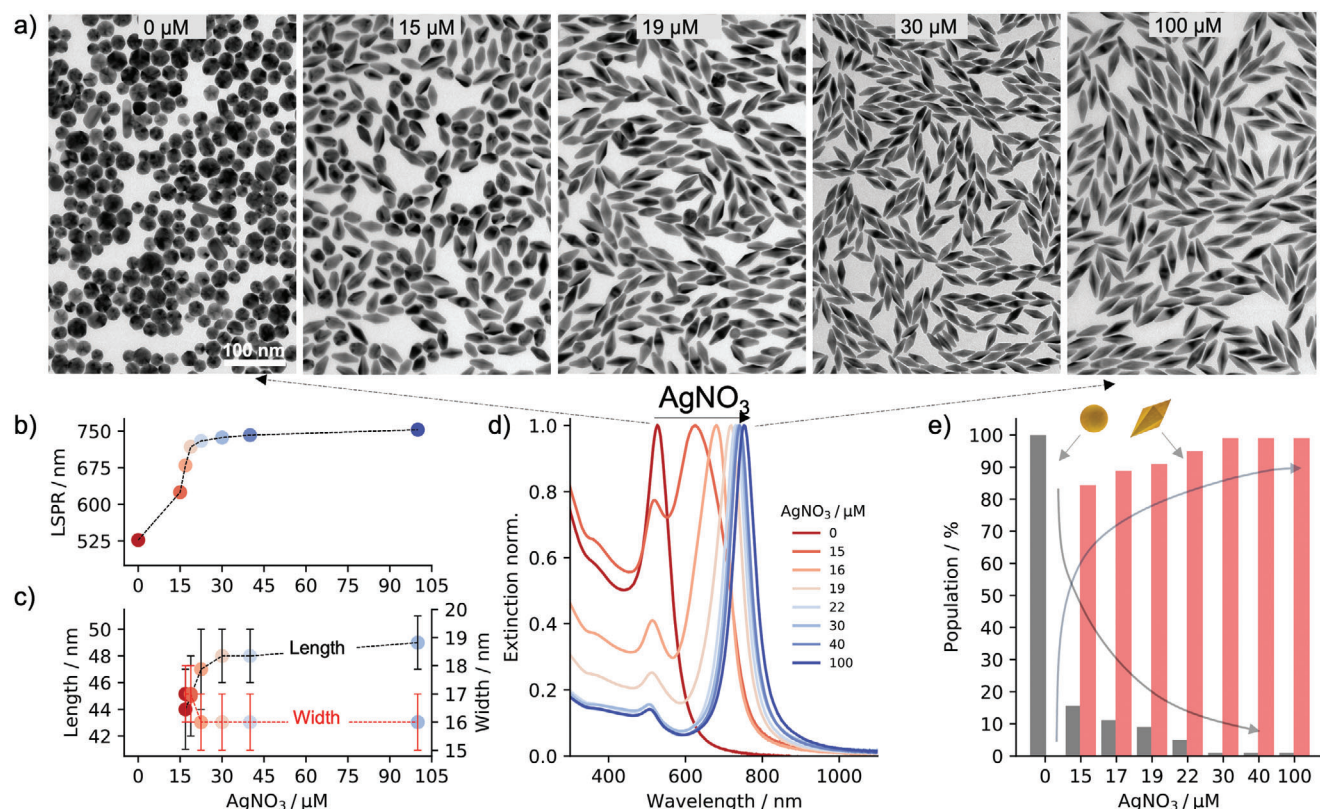


Figure 2. Silver ions concentration controls yield without affecting the dimensions of AuBPs. a) TEM images of AuBPs obtained with increasing AgNO_3 concentration from 0 to 100 μM at $[\text{CTAC}] = 100 \text{ mM}$ and $[\text{CTAB}] = 5 \text{ mM}$. b, c) Effect of AgNO_3 on the position of LSPR (b), length and width (c). d) UV-vis-NIR spectra of AuBPs synthesized at AgNO_3 concentration from 0 to 100 μM as indicated by arrow. e) Change of population of AuBPs and isotropic byproduct as a function of AgNO_3 .

and the elimination of isotropic byproduct are assured, but with limited effect on the aspect ratio.^[29]

To expand the synthetic space, we explored the effect of seed concentration on the dimension of AuBPs at the constant CTAB concentration (5 mM) (Table S4, Supporting Information). With increasing seed concentration from 2.5 to 25 μM , the LSPR blueshifted from 976 to 635 nm (Figure 3d). Accordingly, the length, width, and volume of the particles progressively decreased with increasing seed concentration as the number of nanoparticles in the growth solution increased (Figure 3a,b,c; Table S4 and Figure S8, Supporting Information). Importantly, for the highest seed concentration (25 μM), the dimensions of AuBPs were substantially reduced to 32 ± 2 by $14 \pm 1 \text{ nm}$ (volume $\approx 2300 \text{ nm}^3$), being, to our knowledge, the smallest bipyramids with yield reaching 90% synthesized to date. Note that plasmonic nanocrystals of reduced volume are of interest for therapeutic applications, particularly optical hyperthermia^[58] or optoacoustic imaging^[59,60] requiring a high ratio of absorption-to-extinction cross-section. Since AuBPs exhibit superior thermal stability,^[21] the “minibipyramids” presented here with narrow linewidth are suitable for the mentioned applications and thus outperform gold nanorods.^[58] Finally, we evaluated the possibility of tailoring the aspect ratio in AuBPs through CTAB concentration for the $\text{Au}^{+3}/\text{Au}^0$ molar ratio increasing from 30 to 50 to 67. We observed that regardless of the $\text{Au}^{+3}/\text{Au}^0$ used, the LSPR redshifted with decreasing CTAB concentration (Figure 3e; Tables S5 and S6

and Figures S9 and S10, Supporting Information). These results show that the reasonable combination of CTAB and seed concentrations is a convenient tool for controlling the dimensions of bipyramids.

Typically, an overgrowth of anisotropic nanocrystals (e.g., rods) leads to the loss of shape anisotropy due to the preferential metal reduction on high-index facets, rendering isotropic structures of low-index facets.^[61] On the contrary, the particular structure of bipyramid renders a wide variety of anisotropic morphologies through the overgrowth.^[30] We evaluated the effect of CTAB on the overgrowth of as-prepared AuBPs (66 and 19 nm, Figure S11, Supporting Information) without altering the composition of the growth solution. The concentration of CTAB was increased from 2 to 50 mM while keeping constant the molar ratio of $\text{Au}^{+3}/\text{Au}^0 = 5$, $\text{CTAC} = 100 \text{ mM}$, and $\text{Ag}^+ = 30 \mu\text{M}$ (Table S7, Supporting Information). We observed that upon overgrowth, the initial bipyramidal shape was conserved (Figure 4a; Figure S12, Supporting Information). Similar to the seeded growth from 6 nm spheres, the concentration of CTAB significantly impacted the final dimensions of AuBPs. Upon the increase of CTAB concentration (2–50 mM), the length decreased from 173 ± 4 to $76 \pm 1 \text{ nm}$ while the width increased from 27 ± 2 to $36 \pm 1 \text{ nm}$. The aspect ratio decreased from 6.4 to 2.2 (Figure 4b,d). The position of LSPR of final nanoparticles covered the spectral range from 662 to 1274 nm for the limiting CTAB concentrations of 50 and 2 mM, respectively (Figure 4c). Interestingly, at the lowest CTAB

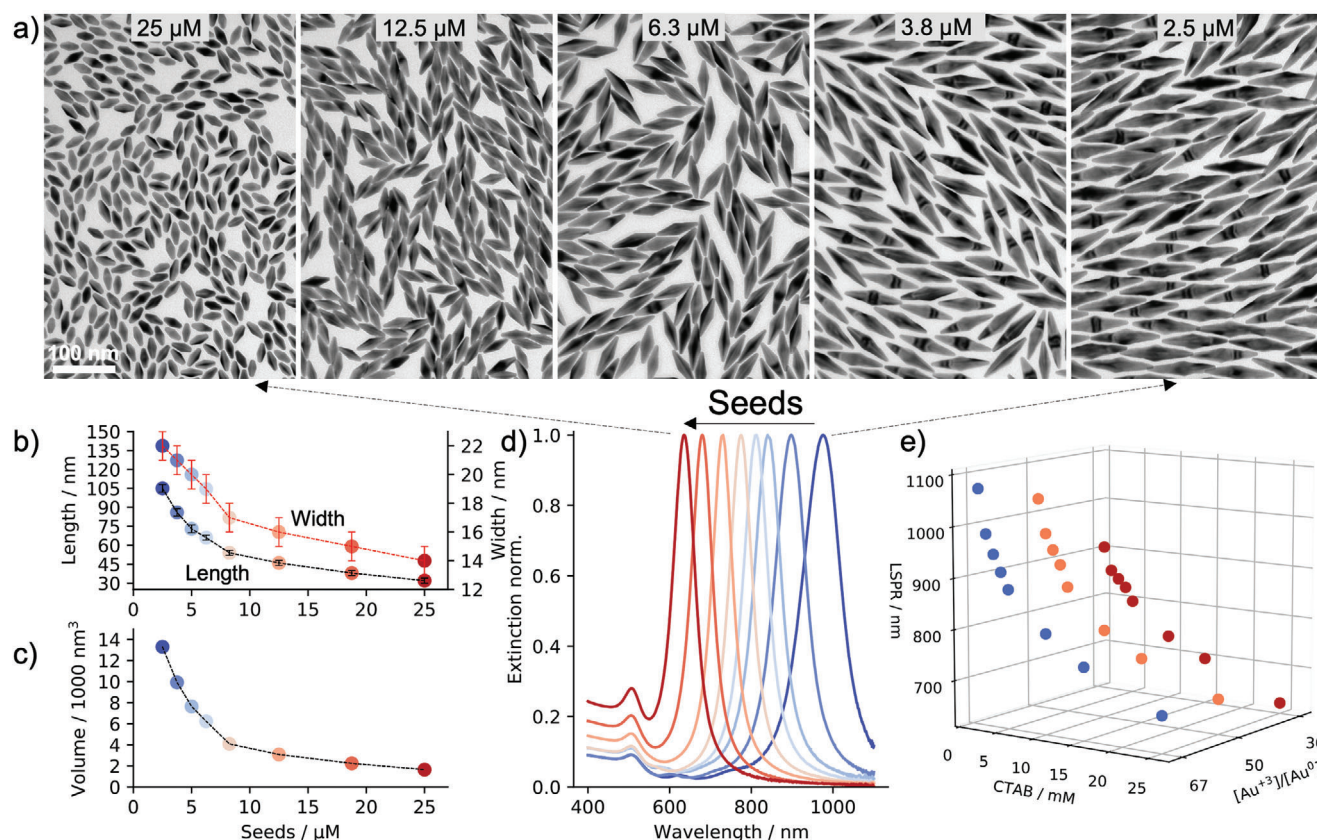


Figure 3. Reducing the volume of AuBPs by tailoring seeds concentration. A) TEM images of AuBPs synthesized with seeds concentration varying from 25 to 2.5 μM , showing size decrease with increasing seeds number. b, c) Change of length and width (b) and volume (c) of final AuBPs as a function of seeds nanoparticles. d) UV-vis-NIR spectra of AuBPs synthesized in the presence of seeds concentration from 2.5 to 25 μM as indicated by the arrow. e) Change of longitudinal LSPR position as a function of CTAB concentration for $\text{Au}^{3+}/\text{Au}^0$ molar ratios from 30, 50, to 67.

concentration (2 mM), we observed the formation of spherical tips, corroborating the results reported by the Weizmann group^[30] (Figure 4a). At low CTAB concentration (CTAC: CTAB = 50), the tips are covered preferentially by $\text{CTA}^+/\text{[AgCl}_2\text{]}^-$ where binding preferences of $\text{CTA}^+/\text{[AgBr}_2\text{]}^-$ over $\text{CTA}^+/\text{[AgCl}_2\text{]}^-$ is minimal. As a result, less protected tips are subjected to gold reduction, rendering a dumbbell-like structure. These results show that the experimental model presented here, in which CTAB is the factor tailoring aspect ratio, enables hierarchical synthesis to reach larger dimensions of bipyramidal nanoparticles. The monodispersed anisotropic nanocrystals are convenient building blocks for programmed self-assembly to obtain crystal habit superlattices similar to their corresponding building blocks.^[62]

The results presented above indicate that the two experimental parameters – CTAB and seed concentration – suffice to tune the dimensions broadly and, thus, the optical properties of AuBPs. We synthesized AuBPs of fixed LSPR at ≈ 750 nm to demonstrate such a possibility while varying nanoparticle volume. The seed concentration was used to vary the volume of final nanoparticles (primary experimental parameter). The concentration of CTAB was used to fine-tune the position of LSPR at the desired wavelength (secondary experimental parameter). By executing a sequence of experiments, we obtained six samples featuring virtually the same optical properties (Figure 5c), varying their volume from 3110 to 19702 nm^3 (Figure 5a,b). The 6-fold change

of volume at fixed LSPR decreases the scattering-to-extinction ratio, making the particles relevant for photo-thermal applications. Note that tailoring volume by seeds and CTAB is valid for the entire spectral range (Figure S13, Supporting Information). Finally, the execution of the study presented here generated 56 high-quality samples. Figure 5d shows the relationship of structural and optical properties (length, width, and position of LSPR) in all samples, where the dimensions of final AuBPs obtained by seeded growth from 6 nm seeds can be substantially expanded by the overgrowth of as-prepared bipyramids. The position of LSPR can be tuned from 600 to ≈ 1300 nm. These results indicate the possibility for further expansion of available dimensions.

To quantify the quality presented here AuBPs, we determined the refractive index sensitivity (RIS) and the Figure of Merit (FOM)^[3,63] for nanocrystals of increasing volume at constant LSPR (≈ 800 nm) in water-glycerol mixtures (0–100% in glycerol, Figure S14, Supporting Information).^[64] We observed that increasing particle volume from 3181 to 11160 nm^3 increases the FOM value from 5.2 to 6.3 because of the drop of FWHM and the increase of RIS with increasing volume of nanoparticles (Figures S15–S17, Supporting Information). Notably, the obtained FOM values are the highest reported for this volume range,^[65] confirming nanocrystals of reasonable quality and excellent candidates for biosensing.

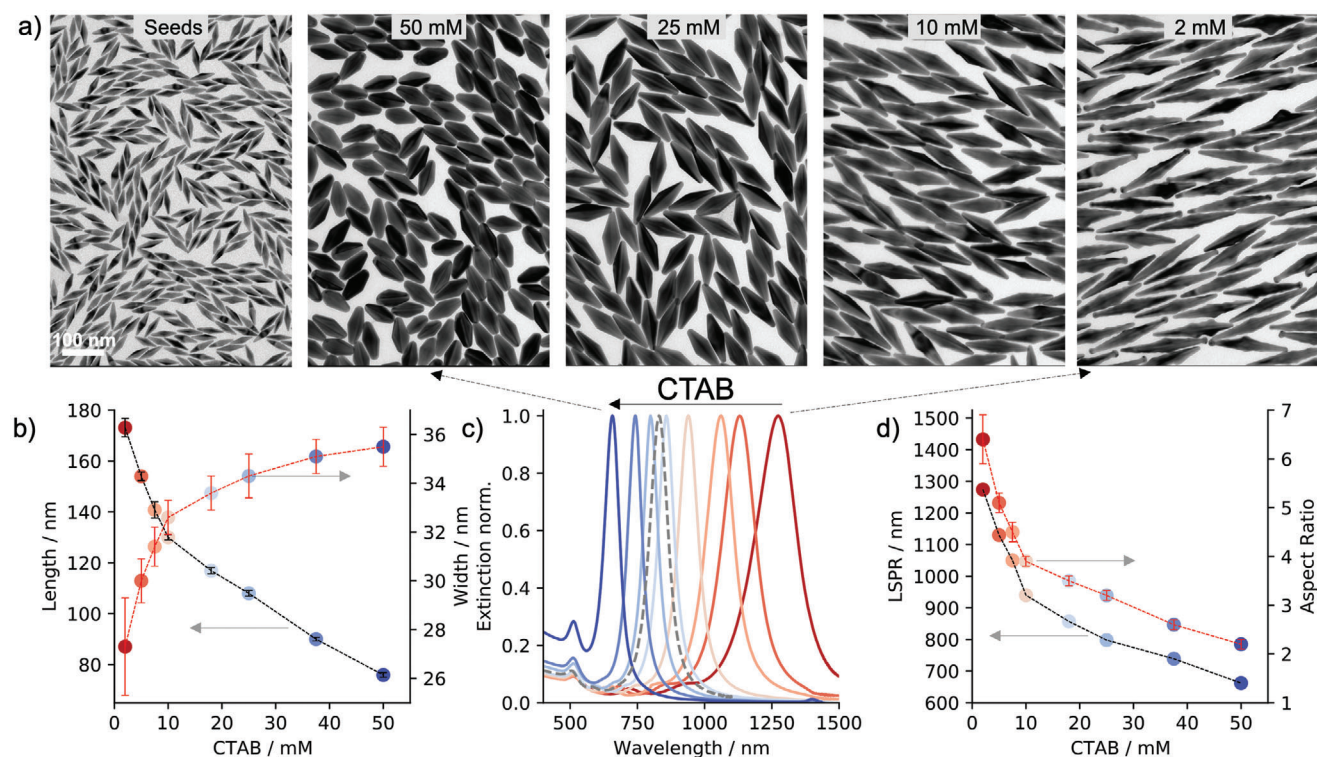


Figure 4. Overgrowth of AuBPs. a) TEM images of initial AuBPs seeds (left) and final AuBPs after overgrowth at varying CTAB concentrations from 2 to 50 mM, showing the change of aspect ratio. b) Length and width of final AuBPs as a function of CTAB concentration. c) UV-vis-NIR spectra of final AuBPs obtained at increasing CTAB concentration as indicated by the arrow. The spectrum of AuBPs seeds is shown in dashed grey. d) Evolution of LSPR position and aspect ratio as a function of CTAB concentration.

It is widely acknowledged that tip curvature is a key factor influencing the position of the plasmon in AuBPs.^[5,66–68] Even a slight increase in tip radius by a few nm can cause a significant blueshift in the plasmon band. The rounding of tips can affect the long-term stability of the LSPR position, potentially compromising targeted applications.^[69] We found that the structural integrity of as-prepared AuBPs is highly dependent on the type of surfactant used for storage. We monitored the optical properties of as-prepared AuBPs (LSPR = 900 nm) dispersed in pure CTAC (10 mM) and pure CTAB (10 mM), and we found that after 30 days of storage at 27 °C, the maximum of LSPR blueshifted by 45 nm in CTAB while remained unchanged in CTAC (Figure S18, Supporting Information). A detailed structural analysis of both samples showed a difference (0.7 nm) in tip curvature and unaltered aspect ratio (Figure S18e, Supporting Information). This change is likely due to the difference in redox potential between AuBr_2^- and AuCl_2^- , leading to the reshaping of the nanoparticles.^[70] Overall, CTAC is a preferred stabilizer for long-term storage of AuBPs, as long as the conservation of the initial optical properties is concerned.

3. Conclusion

We have shown that bromide as a counterion of cationic surfactant is a versatile parameter determining the structural and optical properties of AuBPs. We found an optimal range of CTAB concentration (1–50 mM), allowing for tailoring the position of

the LSPR across a broad spectral range (600–1300 nm). The increase of bromide in the growth solution slows the kinetics of precursor reduction and thus inhibits longitudinal growth, giving rise to a lower aspect ratio. We also found that the concentration of seeds, either thermally-aged spherical nanoparticles or bipyramids, are additional experimental parameters to tailor the final dimensions of AuBPs. The combination of CTAB and seed concentration allowed for varying nanoparticle volumes while keeping the LSPR's position. From a broader perspective, the expanded synthesis space presented here opens up the possibilities for further evaluation of other experimental parameters (e.g., pH^[71]) to ultimately consolidate AuBPs in a wide range of applications.

4. Experimental Section

Synthesis of Gold Seeds: The gold seeds were prepared by fast reduction of gold precursor with NaBH_4 in the presence of a cationic surfactant (CTAC) and citric acid in a 20 mL scintillation vial (see Video S1, Supporting Information). To an aqueous solution of CTAC (10 mL, 50 mM) was added a solution of HAuCl_4 (0.05 mL, 50 mM) followed by stirring for ≈ 3 min until complexation of gold precursor with surfactant (transparent, yellowish solution). Next, citric acid (0.05 mL, 1 M) was added followed by the injection of freshly prepared NaBH_4 (0.25 mL, 25 mM) under vigorous stirring at 20 °C. The solution turned from yellow to brownish immediately. 2 min later, the seed solution was placed in a silicone oil bath and heated at 80 °C for 90 min under stirring. The solution color turned gradually from brown to red, indicating an increase in

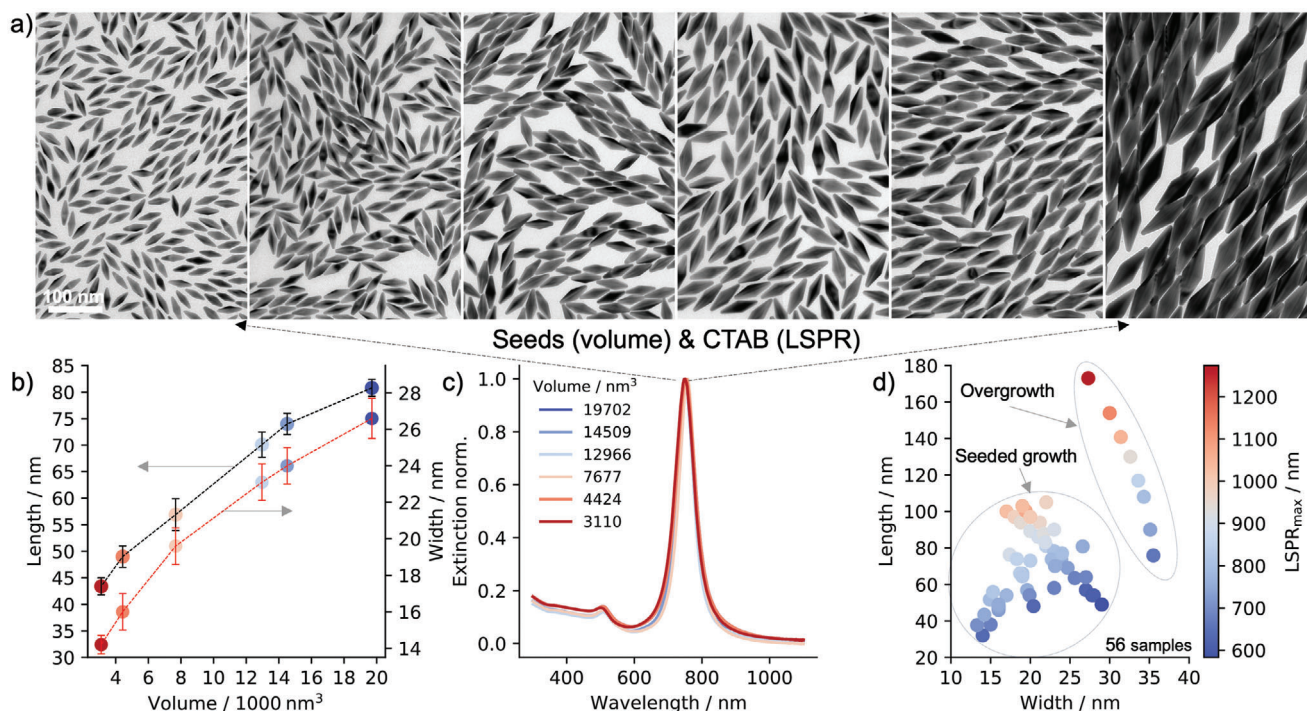


Figure 5. Controlling AuBPs volume at constant LSPR position. a) TEM images of AuBPs featuring volumes ranging from 3110 to 19702 nm³ (from left to right) under constant aspect ratio. b) Change of length and width of nanoparticles in a) showing 6 fold reduction of volume, while keeping a constant aspect ratio. c) UV-vis-NIR spectra of AuBPs featuring constant LSPR at 750 nm shown in a). d) All samples of AuBPs samples (56 in total) presented in length versus width dimension. The points in circles correspond to samples obtained through seeded growth from 6 nm spherical seeds or through overgrowth from as-prepared AuBPs (66 by 19 nm). The color of each point corresponds to the position of LSPR.

nanoparticle size. Finally, the thermally treated seed solution (concentration of gold 0.25 mM) was removed from the oil bath, stored at room temperature, and used without further treatment (see Video S2, Supporting Information).

Synthesis of Gold Bipyramids: To prepare the growth solution, an indicated amount of CTAB was dissolved in an aqueous CTAC solution (20 mL, 100 mM) in a water bath at 35 °C. The mixture was kept under stirring at 35 °C for 10 min. After dissolution of the binary surfactant mixture, HAuCl₄ solution (0.2 mL, 25 mM) was added and the solution was stirred at 35 °C for 5 min to obtain a transparent yellowish solution. Subsequently, AgNO₃ (0.06 mL, 10 mM) and 8-HQL (0.2 mL, 0.4 M in ethanol) were added and the color of the growth solution changed from yellowish to colorless. After 30 s, a desired amount of gold seed solution was added to the growth solution under vigorous stirring. After 15 min, an aliquot of 8-HQL (0.15 mL, 0.4 M in ethanol) was added dropwise, and finally, the mixture was kept at the indicated time in a water bath at 35 °C under stirring (see details in Tables S1–S6 and Video S3, Supporting Information, and Video S4, Supporting Information). Finally, the gold bipyramids dispersions were centrifuged twice (under the conditions as indicated in Supporting Information) to remove the excess of reagents and redispersed in water.

Overgrowth of Gold Bipyramids: To overgrowth gold bipyramids, the same growth solution was used in the formation of gold bipyramids from gold seeds (see details in Table S7, Supporting Information). A dispersion of gold bipyramids with 66 nm in length and, 19 nm in width were used as seeds, with a metallic gold concentration of 5 mM in CTAC 10 mM. Finally, the overgrown gold bipyramids dispersions were centrifuged twice (6000 rpm, 3900 RCF, 30 min) to remove the excess reagents and redispersed in water.

The materials, information on sample analysis and analysis as well as further characterization are provided in Supporting Information.

Statistical Analysis: The data related to the nanoparticles dimensions (length and width) were presented as the mean values, and error bars indi-

cate the standard error of the mean. The sample size for statistical analysis was over 200.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by grant PID2022-141017OB-I00 funded by MCIN/AEI/10.13039/501100011033 and by “ERDF A way of making Europe”. A.S.I. acknowledges financial support from the European Union NextGenerationEU/PRTR-C17.I1, as well as from IKUR Strategy (NanoNeuro) under a collaboration agreement between Ikerbasque Foundation and the Materials Physics Center on behalf of the Department of Education of the Basque Government. The authors thank Anish Rao, Aadesh Mohan Naik, and José Luis Montaña-Priede for the helpful discussion.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

colloidal synthesis, gold bipyramids, plasmonic nanocrystals, seeded growth

Received: September 23, 2024

Revised: October 24, 2024

Published online: November 9, 2024

- [1] T. H. Chow, N. Li, X. Bai, X. Zhuo, L. Shao, J. Wang, *Acc. Chem. Res.* **2019**, 52, 2136.
- [2] Y. Ni, C. Kan, J. Xu, *Phys. Chem. Chem. Phys.* **2022**, 24, 21522.
- [3] X. Zhu, X. Zhuo, Q. Li, Z. Yang, J. Wang, *Adv. Funct. Mater.* **2016**, 26, 341.
- [4] W. Wang, P. Yu, Z. Zhong, X. Tong, T. Liu, Y. Li, E. Ashalley, H. Chen, J. Wu, Z. Wang, *Nanotechnology* **2018**, 29, 355402.
- [5] Q. Ke, L. Chen, B. Fang, Y. Chen, W. Zhang, *Mater. Today Commun.* **2021**, 26, 101953.
- [6] W. Rao, Q. Li, Y. Wang, T. Li, L. Wu, *ACS Nano* **2015**, 9, 2783.
- [7] Q. Li, X. Zhuo, S. Li, Q. Ruan, Q. H. Xu, J. Wang, *Adv. Opt. Mater.* **2015**, 3, 801.
- [8] R. Pardehkhorrām, S. Bonaccorsi, H. Zhu, V. R. Gonçalves, Y. Wu, J. Liu, N. A. Lee, R. D. Tilley, J. J. Gooding, *Chem. Commun.* **2019**, 55, 7707.
- [9] D. Liang, Y. Wang, L. Ma, Y. Liu, R. Fu, H. Liu, Y. Peng, Y. Zhang, C. Wang, B. Jiao, Y. He, *ACS Appl. Nano Mater.* **2022**, 5, 16978.
- [10] M. U. Amin, R. Zhang, L. Li, H. You, J. Fang, *Anal. Chem.* **2021**, 93, 7657.
- [11] C. Kaur, V. Kaur, S. Rai, M. Sharma, T. Sen, *Nanoscale* **2023**, 15, 6170.
- [12] Y. D. Zhang, C. Ma, Y. P. Shi, *Anal. Chim. Acta* **2022**, 1236, 340584.
- [13] P. Zhang, T. Cai, Q. Zhou, G. She, W. Liang, Y. Deng, T. Ning, W. Shi, L. Zhang, C. Zhang, *Nano Lett.* **2022**, 22, 1541.
- [14] S. K. Maji, S. Yu, K. Chung, M. Sekkarapatti Ramasamy, J. W. Lim, J. Wang, H. Lee, D. H. Kim, *ACS Appl. Mater. Interfaces* **2018**, 10, 42068.
- [15] J. Lawless, C. Hrelescu, C. Elliott, L. Peters, N. McEvoy, A. L. Bradley, *ACS Appl. Mater. Interfaces* **2020**, 12, 46406.
- [16] M. Stührenberg, B. Munkhbat, D. G. Baranov, J. Cuadra, A. B. Yankovich, T. J. Antosiewicz, E. Olsson, T. Shegai, *Nano Lett.* **2018**, 18, 5938.
- [17] Q. Ai, F. Sterl, H. Zhang, J. Wang, H. Giessen, *ACS Nano* **2021**, 15, 19409.
- [18] J. H. Lee, Z. Cheglakov, J. Yi, T. M. Cronin, K. J. Gibson, B. Tian, Y. Weizmann, *J. Am. Chem. Soc.* **2017**, 139, 8054.
- [19] O. Shelonchik, N. Lemcoff, R. Shimoni, A. Biswas, E. Yehezkel, D. Yesodi, I. Hod, Y. Weizmann, *Nat. Commun.* **2024**, 15, 1154.
- [20] A. Biswas, N. Lemcoff, O. Shelonchik, D. Yesodi, E. Yehezkel, E. Y. Finestone, A. Upcher, Y. Weizmann, *Nat. Commun.* **2023**, 14, 6355.
- [21] Y. Xu, X. Wang, L. Cheng, Z. Liu, Q. Zhang, *Chem. Eng. J.* **2019**, 378, 122025.
- [22] X. Liu, W. Zhou, T. Wang, S. Miao, S. Lan, Z. Wei, Z. Meng, Q. Dai, H. Fan, *Sci. Rep.* **2023**, 13, 3372.
- [23] X. Chen, Z. Liu, S. G. Parker, X. Zhang, J. J. Gooding, Y. Ru, Y. Liu, Y. Zhou, *ACS Appl. Mater. Interfaces* **2016**, 8, 15857.
- [24] X. Zhu, H. Jia, X. M. Zhu, S. Cheng, X. Zhuo, F. Qin, Z. Yang, J. Wang, *Adv. Funct. Mater.* **2017**, 27, 1700016.
- [25] Y. Ma, X. Zhu, S. Xu, G. He, L. Yao, N. Hu, Y. Su, J. Feng, Y. Zhang, Z. Yang, *Appl. Catal. B Environ.* **2018**, 234, 26.
- [26] N. Lemcoff, N. B. Nechmad, O. Eivgi, E. Yehezkel, O. Shelonchik, R. S. Phatake, D. Yesodi, A. Vaisman, A. Biswas, N. G. Lemcoff, Y. Weizmann, *Nat. Chem.* **2023**, 15, 475.
- [27] J. Lyu, W. Chaabani, E. Modin, A. Chuvin, T. Bizien, F. Smalenburg, M. Impéror-Clerc, D. Constantin, C. Hamon, *Adv. Mater.* **2022**, 34, 2200883.
- [28] Z. Li, Y. Lim, I. Tanriover, W. Zhou, Y. Li, Y. Zhang, K. Aydin, S. C. Glotzer, C. A. Mirkin, *Sci. Adv.* **2024**, 10, eadp3756.
- [29] M. Liu, P. Guyot-Sionnest, *J. Phys. Chem. B* **2005**, 109, 22192.
- [30] J. H. Lee, K. J. Gibson, G. Chen, Y. Weizmann, *Nat. Commun.* **2015**, 6, 7571.
- [31] D. Chateau, A. Liotta, F. Vardcard, J. R. G. Navarro, F. Chaput, J. Lermé, F. Lerouge, S. Parola, *Nanoscale* **2015**, 7, 1934.
- [32] A. Sánchez-Iglesias, N. Winckelmans, T. Altantzis, S. Bals, M. Grzelczak, L. M. Liz-Marzán, *J. Am. Chem. Soc.* **2017**, 139, 107.
- [33] K. Park, S. Biswas, S. Kanel, D. Nepal, R. A. Vaia, *J. Phys. Chem. C* **2014**, 118, 5918.
- [34] J. Zheng, X. Cheng, H. Zhang, X. Bai, R. Ai, L. Shao, J. Wang, *Chem. Rev.* **2021**, 121, 13342.
- [35] K. Cruse, A. Trewartha, S. Lee, Z. Wang, H. Huo, T. He, O. Kononova, A. Jain, G. Ceder, *Sci. Data* **2022**, 9, 234.
- [36] S. Lee, K. Cruse, S. P. Gleason, A. P. Alivisatos, G. Ceder, A. Jain, *ChemRxiv* **2024**, <https://doi.org/10.26434/chemrxiv-2024-ncjlp>.
- [37] H. Chen, L. Shao, T. Ming, Z. Sun, C. Zhao, B. Yang, J. Wang, *Small* **2010**, 6, 2272.
- [38] S. M. Meyer, J. Pettine, D. J. Nesbitt, C. J. Murphy, *J. Phys. Chem. C* **2021**, 125, 16268.
- [39] S. K. Meena, F. Lerouge, P. Baldeck, C. Andraud, M. Garavelli, S. Parola, M. Sulpizi, I. Rivalta, *Nanoscale* **2021**, 13, 15292.
- [40] A. Sánchez-Iglesias, K. Jenkinson, S. Bals, L. M. Liz-Marzán, *J. Phys. Chem. C* **2021**, 125, 23937.
- [41] X. Zhuo, X. Zhu, Q. Li, Z. Yang, J. Wang, *ACS Nano* **2015**, 9, 7523.
- [42] N. R. Jana, L. Gearheart, C. J. Murphy, *J. Phys. Chem. B* **2001**, 105, 4065.
- [43] S. K. Meena, M. Sulpizi, *Langmuir* **2013**, 29, 14954.
- [44] N. Garg, C. Scholl, A. Mohanty, R. Jin, *Langmuir* **2010**, 26, 10271.
- [45] M. Brown, B. J. Wiley, *Chem. Mater.* **2020**, 32, 6410.
- [46] J. Rodríguez-Fernández, J. Pérez-Juste, P. Mulvaney, L. M. Liz-Marzán, *J. Phys. Chem. B* **2005**, 109, 14257.
- [47] J. E. Ortiz-Castillo, R. C. Gallo-Villanueva, M. J. Madou, V. H. Perez-Gonzalez, *Coord. Chem. Rev.* **2020**, 425, 213489.
- [48] J. Pérez-Juste, L. M. Liz-Marzán, S. Carnie, D. Y. N. Chan, P. Mulvaney, *Adv. Funct. Mater.* **2004**, 14, 571.
- [49] Q. N. Nguyen, C. Wang, Y. Shang, A. Janssen, Y. Xia, *Chem. Rev.* **2023**, 123, 3693.
- [50] C. Lofton, W. Sigmund, *Adv. Funct. Mater.* **2005**, 15, 1197.
- [51] M. L. Personick, C. A. Mirkin, *J. Am. Chem. Soc.* **2013**, 135, 18238.
- [52] Z. C. Canbek, R. Cortes-Huerto, F. Testard, O. Spalla, S. Moldovan, O. Ersen, A. Wisnet, G. Wang, J. Goniakowski, C. Noguera, N. Menguy, *Cryst. Growth Des.* **2015**, 15, 3637.
- [53] J. Burgin, I. Florea, J. Majimel, A. Dobri, O. Ersen, M. Tréguer-Delapierre, *Nanoscale* **2012**, 4, 1299.
- [54] P. S. Kumar, I. Pastoriza-Santos, B. Rodríguez-González, F. J. García de Abajo, L. M. Liz-Marzán, *Nanotechnology* **2008**, 19, 015606.
- [55] H. Yuan, C. G. Khoury, H. Hwang, C. M. Wilson, G. A. Grant, T. Vo-Dinh, *Nanotechnology* **2012**, 23, 075102.
- [56] M. R. Langille, M. L. Personick, J. Zhang, C. A. Mirkin, *J. Am. Chem. Soc.* **2012**, 134, 14542.
- [57] B. Nikoobakht, M. A. El-Sayed, *Chem. Mater.* **2003**, 15, 1957.
- [58] H. H. Chang, C. J. Murphy, *Chem. Mater.* **2018**, 30, 1427.
- [59] R. García-Álvarez, L. Chen, A. Nedilko, A. Sánchez-Iglesias, A. Rix, W. Lederle, V. Pathak, T. Lammers, G. von Plessen, K. Kostarelos, L. M. Liz-Marzán, A. J. C. Kuehne, D. N. Chigrin, *ACS Photonics* **2020**, 7, 646.
- [60] Y. Liu, Z. Li, Z. Yin, H. Zhang, Y. Gao, G. Huo, A. Wu, L. Zeng, *ACS Appl. Mater. Interfaces* **2020**, 12, 14866.

- [61] E. Carbó-Argibay, B. Rodríguez-González, J. Pacifico, I. Pastoriza-Santos, J. Pérez-Juste, L. M. Liz-Marzán, *Angew. Chem., Int. Ed.* **2007**, 46, 8983.
- [62] C. Hamon, L. M. Liz-Marzán, *Chem. – Eur. J.* **2015**, 21, 9956.
- [63] K. M. Mayer, J. H. Hafner, *Chem. Rev.* **2011**, 111, 3828.
- [64] H. Chen, X. Kou, Z. Yang, W. Ni, J. Wang, *Langmuir* **2008**, 24, 5233.
- [65] C. Fang, G. Zhao, Y. Xiao, J. Zhao, Z. Zhang, B. Geng, *Sci. Rep.* **2016**, 6, 36706.
- [66] J. L. Montaño-Priede, A. Sánchez-Iglesias, S. A. Mezzasalma, J. Sancho-Parramon, M. Grzelczak, *J. Phys. Chem. Lett.* **2024**, 3914.
- [67] E. Ringe, J. Zhang, M. R. Langille, C. A. Mirkin, L. D. Marks, R. P. V. Duyne, *Nanotechnology* **2012**, 23, 444005.
- [68] J. Marcheselli, D. Chateau, F. Lerouge, P. Baldeck, C. Andraud, S. Parola, S. Baroni, S. Corni, M. Garavelli, I. Rivalta, *J. Chem. Theory Comput.* **2020**, 16, 3807.
- [69] L. Scarabelli, A. Sánchez-Iglesias, J. Pérez-Juste, L. M. Liz-Marzán, *J. Phys. Chem. Lett.* **2015**, 6, 4270.
- [70] A. J. Bard, *Standard Potentials in Aqueous Solution*, Routledge, New York, **2017**.
- [71] X. Zhang, R. Gallagher, D. He, G. Chen, *Chem. Mater.* **2020**, 32, 5626.